Linear Regression

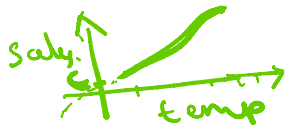
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* It comes under Supervised Learning [make use of both i/p & o/p] (i/p = X, o/p = Y)

* Linear regression is regression algorithm which is used to predict continuous target variable.

* It is used to find linear relationship between dependent (Y) & independent (X) variable by fitting/drawing a straight line.

eg:- Y = "sale" X = "temp"



* It will use straight line equation to find the relationship

$y = mx + c$ → simple linear equation

$y = m_1x_1 + m_2x_2 + m_3x_3 + c$ → multi linear equation for multiple input variables.

where, m = slope
c = intercept.

Assumptions of Linear Regression

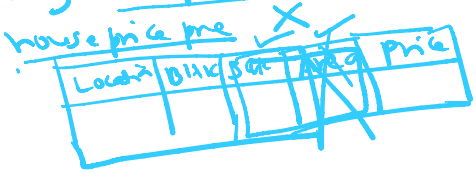
① Linearity - means that there should be linear relationship between X (i/p) & Y (o/p).

* That means there should be high correlation between X & Y.
* Increase of X should lead to increase of Y.

② Data should be normally distributed.

③ There should be little or no multicollinearity → It means that there should not be high correlation among independent variables.

X_1, X_2, X_3, X_4



How Linear Regression works

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X No. of hrs Studied	Y Marks obtained
4	50
7	86
10	92
6	75
8	85
9 (x)	9 88

(1) Linear Regression learns data by drawing straight line.

(2) It uses eqn of straight line $\boxed{y = mx + c}$ to learn data.

(3) We will find value m & c.

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

$\bar{x}$ = mean of X

$\bar{y}$ = mean of Y

x_i = data pair of x

y_i = data pair of y.

$\boxed{m = 7}$, $\boxed{c = 25}$

eqn of str line $y = mx + c$ $\Rightarrow$ $\boxed{y = 7x + 25}$ → best fit line / Regression model.

$y = 9 \times 7 + 25$

$\boxed{y = 88}$

How to check whether the model is best or not?

error/Loss $(y - \hat{y})$

x	actual y	Predicted $\hat{y}$	
4	60	58	2
2	30	31	1
10	92	90	2
		80	5

2	50	90	2
10	92		5
8	85	80	

If error is high, we say model is bad, but if error is less we say, model is good.

To find avg error / Total in Linear Regression, we have some means / approaches :—

① Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum (y_i - \hat{y})^2$$

y_i = actual value
$\hat{y}$ = predicted.
n = Total no. of observation

$$MSE = \frac{2^2 + 1^2 + 2^2 + 5^2}{4} = ?$$

② Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum |y_i - \hat{y}|$$

③ Root Mean Square Error (RMSE)

$$RMSE = \sqrt{MSE}$$

IMP TO NOTE:

* MSE, MAE, RMSE are the error functions which tells about total error made by the model.
* Model is best only if the total error is less.
* Error = actual - predicted.

How to evaluate Model / How to measure performance of model

In Linear regression we can use, R-Square / R2Score

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It tells us as how good ur your model in terms of percentage.

$$R^2 = 1 - \frac{RSS}{TSS}$$

where, RSS → Residual sum of Square.

$$RSS = \sum (y_i - \hat{y})^2$$

$\hat{y}$ → predicted value

y_i → actual value.

TSS → Total sum of Square

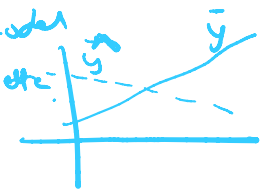
$$TSS = \sum (y_i - \bar{y})^2$$

$\bar{y}$ → average value of y

y_i → actual value.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y})^2}{\sum (y_i - \bar{y})^2}$$

$R^2 = 0.9234$



Your model is 92% better.

R^2 Comparing best fit line with avg line.

i.e It comparing error made by regression line & error made by avg line.

$R^2 < 0$ → bad model

$R^2 = 0$ → bad model

$R^2 \simeq 1$ → Best model.